Decentralizing Agriculture

A shift in the industrial paradigm of food production

Marco Zolla Salazar

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Outline

Land use

- Cropping and animal husbandry cover over one third of the terrestrial land surface [1, p. XVI].
- Agriculture currently drives 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area. [2, p. 4]
- Now, as little as 13% of the ocean and 23% of the land falling under the classification of 'wilderness' [1, p. 206]

Soil degradation and erosion

- Unsustainable agriculture significantly accelerates soil erosion beyond natural rates. [3]
- Soil erosion from agricultural systems causes enormous soil loss globally. [4]
- Human-driven land degradation (including erosion from agriculture) significantly reduces crop productivity. [5]

Pesticide use

- Excessive use and misuse of pesticides result in contamination of surrounding soil and water sources [6]
- Pesticide exposure is associated with "neurological disorders, cancers, reproductive disorders, and respiratory diseases [7]

Fertilizer use

- Nutrient losses from fertilizer use contribute significantly to eutrophication of freshwater and coastal ecosystems [8]
- Nitrogen fertilizer use is associated with elevated nitrate pollution concentrations in agricultural water bodies [9]

Monoculture

- Unsustainable monoculture-based agriculture... drives biodiversity loss, soil degradation, nutrient depletion and contamination of soil and water [10], [11]
- Monoculture agriculture is highly vulnerable to pests compared with plant species-diverse agroecosystems [12]
- Monocultures frequently increase dependence on pesticides contaminating air, water and soil [13]

Produce transportation, storage and disposal

- More than one-third of food is lost or wasted in postharvest operations [14]
- When food is discarded in a landfill and decomposes anaerobically, it yields methane emissions [15]
- Food systems also generate emissions from processing, storage, refrigeration, retail, waste disposal, catering, and transport. [16]

The existential dependency on a few producers

- Four companies account for up to 90 per cent of the global grain trade... concentration runs the risk of reducing the resilience of the global food system. [17]

- Dominant agri-food firms have become too big to feed humanity sustainably and equitably. [18]

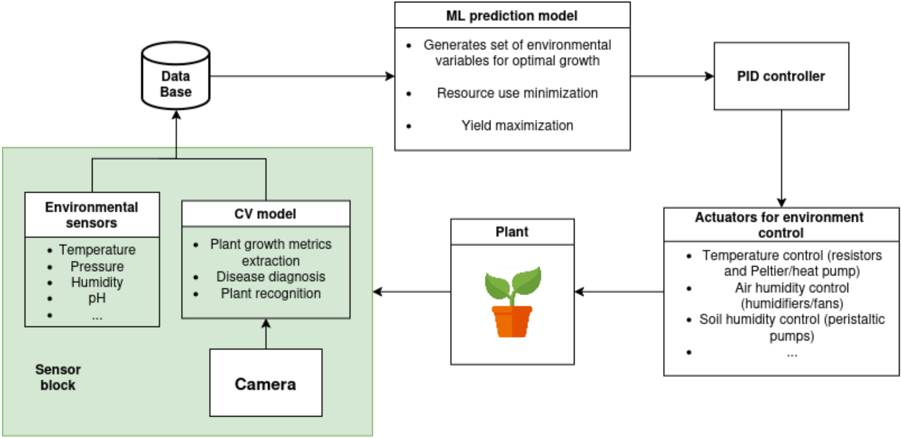
Case study: The Netherlands

- "Dutch greenhouses aren't your run-of-the-mill structures - they're technological marvels." [20]
- The Netherlands, despite its small land area, is the second-largest agricultural exporter in the world by value [19]

Objectives

- Agriculture decentralization
- Optimize production efficiency
- Create a scalable, automatic and widely available solution

Architecture overview



Optimization through collaboration

- Open source hardware, software, models and data
- Guides for personal and commercial use
- Seed sharing communities and markets

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